

Qualification Pack



Cloud Engineer

QP Code: SSC/Q8302

Version: 4.0

NSQF Level: 6.5

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SSC/Q8302: Cloud Engineer

Brief Job Description

Individuals in this job are responsible for designing, maintaining, evaluating, and testing architectures of cloud products and solutions. They formulate enterprise policies and guidelines for cloud implementation and governance. They assist in validating customer proposals and project plans for cloud implementation. They also assess cloud vendors and third-party cloud service providers.

Personal Attributes

Cloud Architects must be able to communicate effectively. They must have strong interpersonal skills, an analytical mindset, and a zeal to learn new things.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [SSC/N8312: Assist in creating proposals and cost models for implementing cloud solutions](#)
2. [SSC/N8311: Validate project plans, costs, and resources required for implementing cloud solutions](#)
3. [SSC/N8310: Define parameters and tools to monitor and manage cloud services](#)
4. [SSC/N8309: Ensure cloud platforms and services are scalable, resilient, and fault-tolerant](#)
5. [SSC/N8308: Design and maintain strong security architecture](#)
6. [SSC/N8307: Analyze, design, and maintain Cloud Engineerure that meets business requirements](#)
7. [SSC/N8306: Translate business-level use cases into technical requirements and specifications](#)
8. [SSC/N8305: Define enterprise cloud guidelines and implement regulatory compliances](#)
9. [SSC/N8304: Assess and screen cloud vendors and third-party service providers](#)
10. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	IT-ITeS
Sub-Sector	Future Skills

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Occupation	Cloud Computing
Country	India
NSQF Level	6.5
Credits	23
Aligned to NCO/ISCO/ISIC Code	NCO-2015/NIL
Minimum Educational Qualification & Experience	PG in any field (Completed 1-year PG.PG in Computer Science/ Information Technology/ Electrical and Electronics Engineering) OR Completed 4 year UG program (UG in Computer Science/ Information Technology/ Electrical and Electronics Engineering) with 1.5 years of experience Relevant Experience in cloud application development or DevOps OR Previous relevant Qualification of NSQF Level (6) with 1.5 years of experience Relevant Experience in cloud application development or DevOps OR Previous relevant Qualification of NSQF Level (5.5) with 3 Years of experience Relevant Experience in cloud application development or DevOps
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NIL
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	07/10/2028
NSQC Approval Date	07/10/2025
Version	4.0
Reference code on NQR	QG-6.5-IT-045912025-V2-NASSCOM
NQR Version	2

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SSC/N8312: Assist in creating proposals and cost models for implementing cloud solutions

Description

This unit is about supporting the preparation of proposals for implementing cloud solutions, including defining the scope, planning execution, estimating costs, and assisting in tracking project outcomes.

Scope

The scope covers the following :

- Define project scope
- Plan project implementation
- Estimate project costs
- Drive project implementation

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** support in compiling technical specifications and solution requirements for the proposal
- PC2.** verify that user and business requirements are accurately documented in the proposal
- PC3.** support in drafting a detailed scope of work and project boundaries for the cloud implementation
- PC4.** support in compiling technical specifications and solution requirements for the proposal
- PC5.** assist in preparing an implementation plan with clear timelines and milestones
- PC6.** identify key drivers influencing project execution such as time, cost, and performance
- PC7.** provide inputs on billing models and consumption tracking for different cloud providers
- PC8.** compile a checklist of implementation activities and dependencies
- PC9.** document resource and licensing requirements necessary for project execution
- PC10.** estimate project costs including labor, licenses, infrastructure, and support services
- PC11.** define Key Performance Indicators (KPIs) to measure project success
- PC12.** forecast business and IT consumption levels to inform capacity planning
- PC13.** assess technical implications of the cloud solution and its compatibility with existing infrastructure

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** standards and procedures related to cloud project proposals
- KU2.** policies and protocols for sharing internal and customer data
- KU3.** documentation formats and reporting protocols used for cloud proposals
- KU4.** stakeholder roles and responsibilities in risk analysis and proposal validation

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- KU5.** relevant templates, checklists, and tools for project planning and cost modeling
- KU6.** technical specification parameters for cloud solution design
- KU7.** concepts related to defining project scope and deliverables
- KU8.** planning methodologies used for implementing cloud-based solutions
- KU9.** criteria used for identifying and monitoring project KPIs
- KU10.** billing models used by leading cloud service providers (e.g., pay-as-you-go, reserved instances)
- KU11.** components and structure of various cloud cost estimation models
- KU12.** budget allocation principles and cost forecasting techniques for cloud projects

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate clearly and professionally with team members and stakeholders
- GS2.** read and interpret technical documents, user requirements, and policy manuals
- GS3.** document information accurately using digital tools and standard formats
- GS4.** plan and prioritize tasks to meet deadlines and project timelines
- GS5.** work effectively as part of a collaborative team
- GS6.** identify and resolve basic problems using structured approaches
- GS7.** make decisions based on data, project goals, and organizational policies
- GS8.** manage time effectively while handling multiple responsibilities
- GS9.** adapt to changes in project scope or priorities
- GS10.** maintain attention to detail while reviewing or drafting documents

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. support in compiling technical specifications and solution requirements for the proposal	3	4	-	2
PC2. verify that user and business requirements are accurately documented in the proposal	3	4	-	2
PC3. support in drafting a detailed scope of work and project boundaries for the cloud implementation	2	3	-	1
PC4. support in compiling technical specifications and solution requirements for the proposal	2	3	-	1
PC5. assist in preparing an implementation plan with clear timelines and milestones	3	4	-	2
PC6. identify key drivers influencing project execution such as time, cost, and performance	3	4	-	2
PC7. provide inputs on billing models and consumption tracking for different cloud providers	3	4	-	2
PC8. compile a checklist of implementation activities and dependencies	2	4	-	2
PC9. document resource and licensing requirements necessary for project execution	1	4	-	1
PC10. estimate project costs including labor, licenses, infrastructure, and support services	2	4	-	1
PC11. define Key Performance Indicators (KPIs) to measure project success	2	4	-	1
PC12. forecast business and IT consumption levels to inform capacity planning	2	4	-	2
PC13. assess technical implications of the cloud solution and its compatibility with existing infrastructure	2	4	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8312
NOS Name	Assist in creating proposals and cost models for implementing cloud solutions
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Cloud Computing
NSQF Level	6.5
Credits	2
Version	3.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

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SSC/N8311: Validate project plans, costs, and resources required for implementing cloud solutions

Description

This unit is about evaluating and verifying project plans, costs, and resources associated with implementing cloud solutions.

Scope

The scope covers the following :

- Validate project plans, costs, and resources required for implementing cloud solutions

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** identify key stakeholders involved in cloud solution implementation, including internal teams, external partners, and end users
- PC2.** document and prioritize stakeholder expectations and business goals related to the cloud initiative
- PC3.** evaluate the feasibility of deliverables in the project plan with respect to scope, timeline, and resource availability
- PC4.** validate timelines and milestone schedules based on task complexity, resource capacity, and organizational goals
- PC5.** review SLAs for cloud services to ensure alignment with defined business and operational objectives
- PC6.** define and confirm human and technical resource requirements for each phase of the cloud project
- PC7.** validate detailed cost estimations including cloud infrastructure, software licenses, operational overhead, and contingency costs
- PC8.** confirm overall project schedule, dependencies, and implementation sequencing across multi-cloud or hybrid platforms
- PC9.** seek approval for the validated project plan from key stakeholders and incorporate feedback received
- PC10.** conduct a detailed risk analysis including technical, financial, and compliance risks, and recommend mitigation strategies
- PC11.** create a risk register and validate the risk management framework with appropriate stakeholders

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** organizational policies, procedures, and guidelines related to IT project governance and planning

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- KU2.** Standard Operating Procedures (SOPs) for collaboration and data exchange across teams and departments
- KU3.** documentation norms and compliance practices for project lifecycle management
- KU4.** roles and responsibilities of internal and external stakeholders involved in cloud project execution
- KU5.** industry-standard tools for project planning, budgeting, and risk assessment (e.g., MS Project, JIRA, Risk Matrix)
- KU6.** methods to evaluate and prioritize stakeholder goals and business outcomes
- KU7.** structure and components of comprehensive project plans (Gantt charts, WBS, RACI matrix, etc.)
- KU8.** definition, negotiation, and enforcement of Service Level Agreements (SLAs) with cloud providers
- KU9.** budgeting and cost-optimization approaches such as FinOps or Total Cost of Ownership (TCO) frameworks
- KU10.** project scheduling concepts including critical path method, resource levelling, and timeboxing
- KU11.** methods for identifying, documenting, and evaluating risks using risk matrices or Monte Carlo simulations
- KU12.** cloud compliance and governance risks including data sovereignty, regulatory adherence (GDPR, HIPAA), and shared responsibility models

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively with internal teams and external stakeholders in both written and verbal formats
- GS2.** interpret technical and project-related documents with accuracy and attention to detail
- GS3.** prepare accurate and clear project documentation, including reports, risk registers, and cost estimates
- GS4.** analyze complex project data and make informed decisions in ambiguous situations
- GS5.** apply time management techniques to prioritize tasks and meet project deadlines
- GS6.** collaborate efficiently with cross-functional teams to meet common objectives
- GS7.** identify potential issues proactively and resolve them using structured problem-solving approaches
- GS8.** review and evaluate work for completeness and alignment with quality standards
- GS9.** exercise sound judgment while balancing stakeholder demands and project constraints
- GS10.** use digital tools and cloud-based platforms for collaboration, documentation, and planning

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. identify key stakeholders involved in cloud solution implementation, including internal teams, external partners, and end users	3	5	-	2
PC2. document and prioritize stakeholder expectations and business goals related to the cloud initiative	3	4	-	1
PC3. evaluate the feasibility of deliverables in the project plan with respect to scope, timeline, and resource availability	3	5	-	2
PC4. validate timelines and milestone schedules based on task complexity, resource capacity, and organizational goals	3	5	-	2
PC5. review SLAs for cloud services to ensure alignment with defined business and operational objectives	3	5	-	2
PC6. define and confirm human and technical resource requirements for each phase of the cloud project	3	4	-	2
PC7. validate detailed cost estimations including cloud infrastructure, software licenses, operational overhead, and contingency costs	3	5	-	2
PC8. confirm overall project schedule, dependencies, and implementation sequencing across multi-cloud or hybrid platforms	3	4	-	2
PC9. seek approval for the validated project plan from key stakeholders and incorporate feedback received	2	4	-	2
PC10. conduct a detailed risk analysis including technical, financial, and compliance risks, and recommend mitigation strategies	2	5	-	2
PC11. create a risk register and validate the risk management framework with appropriate stakeholders	2	4	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8311
NOS Name	Validate project plans, costs, and resources required for implementing cloud solutions
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Cloud Computing
NSQF Level	6.5
Credits	2
Version	3.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

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SSC/N8310: Define parameters and tools to monitor and manage cloud services

Description

This unit is about defining metrics and tools required for monitoring the performance and utilization of cloud resources.

Scope

The scope covers the following :

- Define parameters and tools to monitor and manage cloud services

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** review historical and real-time utilization data to identify trends in consumption of compute, storage, and network resources
- PC2.** interpret cloud resource deployment schemas such as auto-scaling groups, containers, and serverless functions for monitoring scope definition
- PC3.** classify services based on usage patterns to identify underutilized, overutilized, or idle resources
- PC4.** define Key Performance Indicators (KPIs) for monitoring performance, availability, and reliability of cloud-hosted applications and services
- PC5.** select and configure appropriate tools (e.g., CloudWatch, Azure Monitor, Datadog, Prometheus) for monitoring based on cloud platform and service architecture
- PC6.** develop a monitoring plan that includes thresholds, alerts, and automated actions for performance degradation or resource breaches
- PC7.** establish methods to track application-level and infrastructure-level metrics using logs, traces, and telemetry data
- PC8.** create dashboards and reports for usage analytics and present insights to technical and business stakeholders
- PC9.** integrate Artificial Intelligence (AI)/Machine Learning (ML) tools into the monitoring ecosystem to enable predictive analytics and intelligent alerting
- PC10.** configure automated anomaly detection using AI/ML algorithms to improve incident response time

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** guidelines and governance policies for resource provisioning and utilization
- KU2.** data-sharing and privacy policies relevant to operational monitoring
- KU3.** documentation standards for logging system metrics, incidents, and monitoring parameters

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- KU4.** roles of relevant stakeholders (e.g., DevOps, security, infrastructure, finance) in performance and cost monitoring
- KU5.** common monitoring templates and dashboards used across major cloud platforms
- KU6.** different resource deployment schemas such as virtual machines, containers, serverless, and microservices
- KU7.** characteristics of cloud resource utilization including elasticity, burst scaling, and shared usage
- KU8.** techniques to define and monitor KPIs for performance, availability, and fault tolerance
- KU9.** tools and frameworks for monitoring usage and performance (e.g., Grafana, ELK stack, Splunk)
- KU10.** tools for anomaly detection, observability, and alert management
- KU11.** principles of AI/ML-based observability, including supervised/unsupervised anomaly detection and AI log analysis
- KU12.** tools enabling intelligent monitoring such as AIOps platforms (e.g., Amazon DevOps Guru, Azure Monitor Insights)
- KU13.** industry practices for cloud cost optimization and performance tuning through monitoring
- KU14.** visualization and reporting tools used to communicate monitoring data effectively

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate clearly with team members and stakeholders to discuss system health and utilization
- GS2.** interpret technical documents, usage logs, and monitoring outputs accurately
- GS3.** document monitoring configurations, thresholds, and tool setup with precision
- GS4.** apply critical thinking to interpret usage trends and anticipate performance bottlenecks
- GS5.** use time management to schedule monitoring tasks and generate reports on time
- GS6.** work collaboratively across DevOps, cloud, and business teams
- GS7.** evaluate trade-offs between cost, performance, and resource utilization
- GS8.** select the appropriate monitoring tools based on system architecture and requirements
- GS9.** solve monitoring-related configuration or integration issues systematically
- GS10.** stay organized while managing multiple cloud environments and toolsets

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. review historical and real-time utilization data to identify trends in consumption of compute, storage, and network resources	3	5	-	2
PC2. interpret cloud resource deployment schemas such as auto-scaling groups, containers, and serverless functions for monitoring scope definition	3	5	-	2
PC3. classify services based on usage patterns to identify underutilized, overutilized, or idle resources	3	5	-	2
PC4. define Key Performance Indicators (KPIs) for monitoring performance, availability, and reliability of cloud-hosted applications and services	3	5	-	2
PC5. select and configure appropriate tools (e.g., CloudWatch, Azure Monitor, Datadog, Prometheus) for monitoring based on cloud platform and service architecture	3	5	-	2
PC6. develop a monitoring plan that includes thresholds, alerts, and automated actions for performance degradation or resource breaches	3	5	-	2
PC7. establish methods to track application-level and infrastructure-level metrics using logs, traces, and telemetry data	3	5	-	2
PC8. create dashboards and reports for usage analytics and present insights to technical and business stakeholders	3	5	-	2
PC9. integrate Artificial Intelligence (AI)/Machine Learning (ML) tools into the monitoring ecosystem to enable predictive analytics and intelligent alerting	3	5	-	2
PC10. configure automated anomaly detection using AI/ML algorithms to improve incident response time	3	5	-	2

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8310
NOS Name	Define parameters and tools to monitor and manage cloud services
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Cloud Computing
NSQF Level	6.5
Credits	2
Version	3.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

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SSC/N8309: Ensure cloud platforms and services are scalable, resilient, and fault-tolerant

Description

This unit is about ensuring that cloud platforms and services are resilient and fault-tolerant and respond to scalability requirements.

Scope

The scope covers the following :

- Ensure cloud platforms and services are scalable, resilient, and fault-tolerant

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** define key performance metrics for measuring the functionality, scalability, latency, and reliability of cloud-based systems and applications
- PC2.** perform systematic testing of cloud platforms and services to validate responsiveness, scalability, fault recovery, and performance thresholds
- PC3.** identify performance bottlenecks and root causes affecting scalability, reliability, and availability
- PC4.** evaluate the adaptability and modifiability of the cloud solution architecture to accommodate changing requirements
- PC5.** assess portability of the cloud solution across multiple environments (hybrid, multi-cloud) to ensure cross-platform compatibility
- PC6.** identify and assess potential internal and external security threats to cloud systems and data
- PC7.** integrate essential security mechanisms such as IAM, encryption, and authentication protocols in cloud infrastructure
- PC8.** develop and implement disaster recovery and incident response plans to ensure business continuity
- PC9.** integrate cloud-native frameworks and patterns (e.g., microservices, containerization) to design scalable cloud solutions
- PC10.** identify and implement auto-scaling, load balancing, and orchestration services for dynamic demand management
- PC11.** utilize analytics and monitoring tools to analyze performance data and inform capacity planning decisions
- PC12.** apply AI-based scaling strategies by using predictive analytics for workload forecasting and dynamic provisioning
- PC13.** validate fault-tolerance using AI-simulated stress tests and scenario-based modeling
- PC14.** schedule and execute periodic system and platform upgrades in line with organizational and technological advancements
- PC15.** validate that upgrades enhance or maintain required levels of security, performance, and scalability

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Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** standards and policies related to testing, security, and performance monitoring of cloud systems
- KU2.** guidelines for documenting performance issues, security incidents, and upgrade logs
- KU3.** roles and responsibilities for defining performance metrics and assessing system health
- KU4.** industry standards and tools for load testing, security testing, and fault tolerance (e.g., Apache JMeter, Chaos Monkey)
- KU5.** metrics for evaluating system behavior under load and failure conditions
- KU6.** techniques to assess modifiability, compatibility, and portability across environments
- KU7.** common vulnerabilities and threat vectors affecting cloud environments
- KU8.** global security standards and regulatory frameworks (e.g., ISO/IEC 27017, ISO/IEC 27018, NIST)
- KU9.** methods to define and apply mitigation strategies and recovery plans
- KU10.** architecture frameworks (e.g., AWS Well-Architected, Google Cloud Engineer Framework) that support scalability and resilience
- KU11.** use of monitoring and analytics solutions (e.g., Prometheus, Grafana, CloudWatch) to inform scaling decisions
- KU12.** system upgrade methodologies and tools (e.g., rolling updates, blue-green deployment)
- KU13.** techniques for applying AI to predictive scaling and auto-remediation
- KU14.** AI-based simulation tools for testing resilience and fault-tolerance in cloud environments

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** interpret and follow cloud strategy guidelines, operational plans, and upgrade schedules
- GS2.** communicate effectively with cross-functional teams and external service providers
- GS3.** analyze impact of system issues and proposed changes on business outcomes
- GS4.** review work output and troubleshoot issues with accuracy and precision
- GS5.** apply logical reasoning to resolve performance and security issues
- GS6.** ensure consistency and reliability in service delivery and user experience
- GS7.** plan tasks and prioritize activities to meet service level objectives (SLOs) and business needs
- GS8.** apply structured problem-solving methods in testing, scaling, or fault resolution scenarios

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. define key performance metrics for measuring the functionality, scalability, latency, and reliability of cloud-based systems and applications	2	3	-	2
PC2. perform systematic testing of cloud platforms and services to validate responsiveness, scalability, fault recovery, and performance thresholds	2	2	-	1
PC3. identify performance bottlenecks and root causes affecting scalability, reliability, and availability	2	5	-	1
PC4. evaluate the adaptability and modifiability of the cloud solution architecture to accommodate changing requirements	2	4	-	1
PC5. assess portability of the cloud solution across multiple environments (hybrid, multi-cloud) to ensure cross-platform compatibility	2	3	-	2
PC6. identify and assess potential internal and external security threats to cloud systems and data	2	3	-	1
PC7. integrate essential security mechanisms such as IAM, encryption, and authentication protocols in cloud infrastructure	2	5	-	1
PC8. develop and implement disaster recovery and incident response plans to ensure business continuity	2	2	-	1
PC9. integrate cloud-native frameworks and patterns (e.g., microservices, containerization) to design scalable cloud solutions	2	3	-	2
PC10. identify and implement auto-scaling, load balancing, and orchestration services for dynamic demand management	2	3	-	2

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. utilize analytics and monitoring tools to analyze performance data and inform capacity planning decisions	2	3	-	1
PC12. apply AI-based scaling strategies by using predictive analytics for workload forecasting and dynamic provisioning	2	4	-	1
PC13. validate fault-tolerance using AI-simulated stress tests and scenario-based modeling	2	4	-	1
PC14. schedule and execute periodic system and platform upgrades in line with organizational and technological advancements	2	3	-	2
PC15. validate that upgrades enhance or maintain required levels of security, performance, and scalability	2	3	-	1
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8309
NOS Name	Ensure cloud platforms and services are scalable, resilient, and fault-tolerant
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Cloud Computing
NSQF Level	6.5
Credits	2
Version	3.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

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SSC/N8308: Design and maintain strong security architecture

Description

This unit is about creating and maintaining a cloud security architecture that addresses security aspects of the business requirements.

Scope

The scope covers the following :

- Design and maintain strong security architecture

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** identify and analyze security requirements based on business needs, customer specifications, and organizational standards
- PC2.** determine and document security needs related to compliance, data privacy, and regulatory standards for cloud solutions
- PC3.** plan and design robust security architectures that fulfill the identified security requirements, considering all aspects of cloud computing
- PC4.** identify and implement recognized security standards, such as ISO/IEC 27017, ISO/IEC 27018, and other relevant protocols, for the cloud solution
- PC5.** implement security measures across all layers of the cloud solution (e.g., compute, network, storage, identity, etc.)
- PC6.** integrate critical security safeguards such as routers, firewalls, intrusion detection/prevention systems (IDS/IPS), Identity and Access Management (IAM) tools, and encryption to protect cloud assets
- PC7.** ensure the security architecture enforces strong authentication and access controls, allowing only authorized users to access sensitive cloud resources
- PC8.** maintain and continuously improve the security architecture by updating it in response to emerging threats and technological advancements
- PC9.** enforce and monitor compliance with organizational security policies, standards, and best practices
- PC10.** ensure that the architecture supports ongoing security operations, including threat and vulnerability management, incident response, and audit processes
- PC11.** develop and implement incident response procedures and monitoring processes to detect, respond to, and recover from security and privacy incidents
- PC12.** conduct security tests, such as vulnerability scanning, penetration testing, and risk assessments, to ensure the security architecture meets all requirements and behaves as expected under different scenarios
- PC13.** validate that security controls and safeguards are correctly implemented and operational across the solution architecture
- PC14.** identify and mitigate any security gaps found through testing or regular monitoring

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PC15. document findings and recommend actions for improving the security posture based on test results and audits

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** policies, procedures, and guidelines for designing, implementing, and maintaining security architectures
- KU2.** data-sharing policies and regulatory compliance requirements related to cloud solutions and security
- KU3.** documentation procedures for security architectures, including authorization protocols, security policies, and risk assessments
- KU4.** stakeholders involved in defining security requirements, including compliance officers, data privacy experts, and the IT security team
- KU5.** different security frameworks, templates, and tools available for designing and implementing secure Cloud Engineerures
- KU6.** the process of defining security requirements based on business, regulatory, and technical needs
- KU7.** different types of security architectures (e.g., perimeter security, zero-trust architecture, etc.) and their applications in cloud environments
- KU8.** various security standards and protocols, including those related to data protection (e.g., ISO/IEC 27017, ISO/IEC 27018, NIST, GDPR)
- KU9.** the range of security safeguards, including encryption, firewalls, access controls, and IAM solutions, and how to apply them to secure cloud infrastructure
- KU10.** data governance and privacy standards, including monitoring and auditing data access, storage, and transmission
- KU11.** access control models (e.g., RBAC, ABAC) and how to enforce them in Cloud Engineerures
- KU12.** security testing methodologies, including threat analysis, vulnerability scanning, penetration testing, and risk assessment tools
- KU13.** Strategies to scale security architectures and handle large-scale, dynamic cloud environments

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** seek clarification and guidance from relevant experts or teams when needed
- GS2.** collaborate effectively with cross-functional teams, including security, IT, and compliance personnel
- GS3.** analyze how architectural and design decisions impact organizational security, operations, and compliance
- GS4.** use data and metrics to assess the security effectiveness of systems and processes
- GS5.** communicate complex security concepts and risks clearly to stakeholders at all levels of the organization

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- GS6.** ensure that work products, including security architectures and implementations, are accurate, comprehensive, and free of errors
- GS7.** exercise sound judgment to resolve security-related issues and make decisions that align with business goals
- GS8.** deliver consistent, high-quality service in securing cloud systems and mitigating risks to cloud-based resources
- GS9.** organize and prioritize tasks to meet deadlines and security objectives
- GS10.** apply systematic problem-solving approaches to security challenges and incidents

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. identify and analyze security requirements based on business needs, customer specifications, and organizational standards	2	3	-	1
PC2. determine and document security needs related to compliance, data privacy, and regulatory standards for cloud solutions	2	3	-	1
PC3. plan and design robust security architectures that fulfill the identified security requirements, considering all aspects of cloud computing	2	5	-	2
PC4. identify and implement recognized security standards, such as ISO/IEC 27017, ISO/IEC 27018, and other relevant protocols, for the cloud solution	2	5	-	2
PC5. implement security measures across all layers of the cloud solution (e.g., compute, network, storage, identity, etc.)	2	4	-	2
PC6. integrate critical security safeguards such as routers, firewalls, intrusion detection/prevention systems (IDS/IPS), Identity and Access Management (IAM) tools, and encryption to protect cloud assets	2	4	-	2
PC7. ensure the security architecture enforces strong authentication and access controls, allowing only authorized users to access sensitive cloud resources	2	3	-	1
PC8. maintain and continuously improve the security architecture by updating it in response to emerging threats and technological advancements	2	2	-	1
PC9. enforce and monitor compliance with organizational security policies, standards, and best practices	2	3	-	1
PC10. ensure that the architecture supports ongoing security operations, including threat and vulnerability management, incident response, and audit processes	2	2	-	1

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. develop and implement incident response procedures and monitoring processes to detect, respond to, and recover from security and privacy incidents	2	3	-	1
PC12. conduct security tests, such as vulnerability scanning, penetration testing, and risk assessments, to ensure the security architecture meets all requirements and behaves as expected under different scenarios	2	4	-	1
PC13. validate that security controls and safeguards are correctly implemented and operational across the solution architecture	2	3	-	2
PC14. identify and mitigate any security gaps found through testing or regular monitoring	2	3	-	1
PC15. document findings and recommend actions for improving the security posture based on test results and audits	2	3	-	1
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8308
NOS Name	Design and maintain strong security architecture
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Cloud Computing
NSQF Level	6.5
Credits	3
Version	3.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

SSC/N8307: Analyze, design, and maintain Cloud Engineerure that meets business requirements

Description

This unit is about designing and maintaining secure cloud architecture that addresses business requirements.

Scope

The scope covers the following :

- Analyze, design, and maintain Cloud Engineerure that meets business requirements

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** assess organizational goals, existing IT infrastructure, and business requirements for the cloud solution
- PC2.** identify a suitable cloud delivery model (e.g., IaaS, PaaS, SaaS, hybrid, multi-cloud) for the solution
- PC3.** conduct feasibility and impact analysis of potential architecture options
- PC4.** define RAID (Risks, Assumptions, Issues, Dependencies) catalog aligned with project objectives
- PC5.** select and apply relevant Cloud Engineerure frameworks (e.g., Cloud Native Computing Foundation, AWS Well-Architected Framework)
- PC6.** design architecture that aligns with business and network requirements across all cloud layers
- PC7.** define deployment schemas for cloud resources and applications
- PC8.** design AI-ready Cloud Engineerures that support GPU/TPU-based computing, scalable storage, and high-throughput networking
- PC9.** incorporate AI-driven services (e.g., image recognition, predictive analytics, NLP) to meet business use cases
- PC10.** incorporate cloud-native principles, scalability, and resilience into the architecture
- PC11.** define and document disaster recovery strategies and high availability plans
- PC12.** establish cloud security policies, standards, and best practices
- PC13.** assess security capabilities of selected cloud service providers
- PC14.** conduct security risk assessments and implement a risk mitigation strategy
- PC15.** define Identity and Access Management (IAM) and role-based access control (RBAC) policies
- PC16.** implement necessary security controls across the solution architecture
- PC17.** document and maintain architecture components and design decisions
- PC18.** review solution design periodically to ensure performance, scalability, and compliance
- PC19.** monitor industry trends and changes in cloud technology standards
- PC20.** update solution architecture accordingly and communicate changes to relevant stakeholders

Qualification Pack

PC21. define and manage a framework for architecture monitoring, evaluation, and reporting

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** policies, procedures, and governance models for IT and cloud solutions
- KU2.** data governance, privacy, and sharing policies
- KU3.** documentation standards and communication protocols for solution architecture
- KU4.** key stakeholders and collaboration roles in the architecture design process
- KU5.** relevant tools, platforms, and templates for Cloud Engineerure design
- KU6.** characteristics and use cases of cloud delivery models (IaaS, PaaS, SaaS, hybrid, multi-cloud)
- KU7.** methods for defining and managing cloud deployment schemas
- KU8.** concepts of cloud-native architecture, containers, and microservices
- KU9.** best practices for disaster recovery and fault tolerance
- KU10.** common Cloud Engineerure frameworks and standards
- KU11.** AI/ML service stack and capabilities of major cloud providers (e.g., AWS SageMaker, Azure ML, Google AI Platform)
- KU12.** design considerations for building and deploying machine learning pipelines in the cloud
- KU13.** RAID catalog methodology and documentation
- KU14.** information security standards (e.g., ISO/IEC 27001, NIST) and cloud security frameworks
- KU15.** identity and access control mechanisms in the cloud
- KU16.** cloud performance monitoring and architecture evaluation tools

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively with stakeholders using oral and written forms
- GS2.** interpret and follow instructions, standard operating procedures, and guidelines
- GS3.** work independently as well as collaboratively in cross-functional teams
- GS4.** evaluate technical solutions based on business impact and requirements
- GS5.** verify the completeness and accuracy of solution architecture documents
- GS6.** apply critical thinking to resolve conflicts and architectural trade-offs
- GS7.** ensure high-quality service delivery in line with customer expectations
- GS8.** evaluate own and team performance against key performance indicators
- GS9.** organize and prioritize tasks to meet project deadlines
- GS10.** troubleshoot and resolve technical challenges through structured problem-solving

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. assess organizational goals, existing IT infrastructure, and business requirements for the cloud solution	1	3	-	1
PC2. identify a suitable cloud delivery model (e.g., IaaS, PaaS, SaaS, hybrid, multi-cloud) for the solution	2	2	-	1
PC3. conduct feasibility and impact analysis of potential architecture options	2	3	-	1
PC4. define RAID (Risks, Assumptions, Issues, Dependencies) catalog aligned with project objectives	1	2	-	1
PC5. select and apply relevant Cloud Engineering frameworks (e.g., Cloud Native Computing Foundation, AWS Well-Architected Framework)	2	5	-	1
PC6. design architecture that aligns with business and network requirements across all cloud layers	2	3	-	1
PC7. define deployment schemas for cloud resources and applications	2	2	-	1
PC8. design AI-ready Cloud Engineering frameworks that support GPU/TPU-based computing, scalable storage, and high-throughput networking	2	2	-	1
PC9. incorporate AI-driven services (e.g., image recognition, predictive analytics, NLP) to meet business use cases	2	2	-	1
PC10. incorporate cloud-native principles, scalability, and resilience into the architecture	1	1	-	1
PC11. define and document disaster recovery strategies and high availability plans	1	2	-	1
PC12. establish cloud security policies, standards, and best practices	2	2	-	1

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. assess security capabilities of selected cloud service providers	1	3	-	1
PC14. conduct security risk assessments and implement a risk mitigation strategy	1	2	-	1
PC15. define Identity and Access Management (IAM) and role-based access control (RBAC) policies	2	3	-	1
PC16. implement necessary security controls across the solution architecture	1	2	-	1
PC17. document and maintain architecture components and design decisions	1	2	-	1
PC18. review solution design periodically to ensure performance, scalability, and compliance	1	3	-	1
PC19. monitor industry trends and changes in cloud technology standards	1	2	-	1
PC20. update solution architecture accordingly and communicate changes to relevant stakeholders	1	2	-	1
PC21. define and manage a framework for architecture monitoring, evaluation, and reporting	1	2	-	-
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8307
NOS Name	Analyze, design, and maintain Cloud Engineerure that meets business requirements
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Cloud Computing
NSQF Level	6.5
Credits	3
Version	3.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

SSC/N8306: Translate business-level use cases into technical requirements and specifications

Description

This unit is about gathering business requirements and translating them into technical requirements and specifications for relevant stakeholders.

Scope

The scope covers the following :

- Map requirements

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** identify and define functional and non-functional requirements from clients, internal departments, or organizational needs
- PC2.** document business requirements using standard tools, templates, and formats
- PC3.** collaborate with business stakeholders to ensure clarity, completeness, and accuracy of use cases and requirements
- PC4.** convert business requirements into detailed technical and IT specifications, including data, application, and integration needs
- PC5.** ensure alignment of technical specifications with organizational architecture, systems, and standards
- PC6.** validate functional and technical requirements with relevant internal and external stakeholders
- PC7.** refine and update requirements based on feedback from business analysts, developers, and clients
- PC8.** finalize and communicate approved requirements to relevant technical teams and stakeholders

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** policies, standards, and best practices for defining and documenting requirements
- KU2.** procedures for data collection, sharing, validation, and retention
- KU3.** governance models for managing and approving business and technical specifications
- KU4.** stakeholders involved in requirement analysis, validation, and decision-making processes
- KU5.** tools, templates, and platforms used for documenting and managing requirements (e.g., JIRA, Confluence, Visio)
- KU6.** principles of functional and non-functional requirement definition
- KU7.** how to derive technical requirements from business use cases

Qualification Pack

KU8. techniques for validating and refining requirements through stakeholder interactions

KU9. risks of misinterpreting or under-specifying requirements and how to mitigate them

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. communicate effectively with business and technical teams, both verbally and in writing

GS2. ask relevant questions to clarify ambiguous or conflicting requirements

GS3. listen actively to capture business needs and concerns

GS4. analyze the business impact of proposed changes and technical solutions

GS5. ensure that documentation is complete, structured, and error-free

GS6. foster positive relationships with customers and stakeholders to build trust

GS7. collaborate effectively in cross-functional teams and client-facing environments

GS8. organize and prioritize tasks to meet deadlines in requirement-gathering and analysis phases

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. identify and define functional and non-functional requirements from clients, internal departments, or organizational needs	4	7	-	3
PC2. document business requirements using standard tools, templates, and formats	4	7	-	2
PC3. collaborate with business stakeholders to ensure clarity, completeness, and accuracy of use cases and requirements	4	5	-	2
PC4. convert business requirements into detailed technical and IT specifications, including data, application, and integration needs	4	7	-	3
PC5. ensure alignment of technical specifications with organizational architecture, systems, and standards	3	6	-	2
PC6. validate functional and technical requirements with relevant internal and external stakeholders	4	6	-	3
PC7. refine and update requirements based on feedback from business analysts, developers, and clients	4	6	-	3
PC8. finalize and communicate approved requirements to relevant technical teams and stakeholders	3	6	-	2
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8306
NOS Name	Translate business-level use cases into technical requirements and specifications
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Cloud Computing
NSQF Level	6.5
Credits	3
Version	4.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

SSC/N8305: Define enterprise cloud guidelines and implement regulatory compliances

Description

This unit is about creating standard protocols and ensuring compliance with regulatory standards for the successful implementation of cloud architecture in the organization.

Scope

The scope covers the following :

- Define enterprise cloud guidelines and implement regulatory compliances

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** analyze organizational business strategies, goals, and IT infrastructure requirements to determine cloud adoption needs
- PC2.** identify and classify all cloud-deployed and on-premise assets based on sensitivity, criticality, and compliance impact
- PC3.** assess integration and interoperability of hybrid/multi-cloud and on-premise environments with a focus on data flow, latency, and security
- PC4.** define programming, infrastructure, and design standards for developing, deploying, and maintaining scalable and secure cloud solutions
- PC5.** establish enterprise-wide standards for provisioning, scaling, and decommissioning of cloud resources
- PC6.** develop role-based access control (RBAC) policies and identity management frameworks for cloud environments
- PC7.** design performance monitoring standards, incident management workflows, and SLAs for cloud infrastructure
- PC8.** create guidelines for contributing to and managing open-source software, as relevant to the enterprise's cloud operations
- PC9.** identify applicable legal, regulatory, and industry-specific standards (e.g., PCI DSS, HIPAA, ISO 27001, GDPR, SOC 2)
- PC10.** implement governance frameworks and internal controls to ensure compliance with defined policies and regulatory standards
- PC11.** monitor, audit, and continuously improve compliance adherence using automated tools and manual checkpoints

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** policies and protocols for IT governance, data sharing, security, and resource usage

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- KU2.** standards and procedures for defining and maintaining IT architecture and cloud-related policies
- KU3.** responsibilities of stakeholders involved in developing and reviewing enterprise IT policies
- KU4.** tools and templates used for documenting policies, standards, and compliance frameworks (e.g., policy frameworks, audit trails)
- KU5.** cloud service models (IaaS, PaaS, SaaS) and deployment models (private, public, hybrid, multi-cloud)
- KU6.** best practices in secure Cloud Engineering, resource tagging, usage metering, and service orchestration
- KU7.** techniques to implement RBAC, multi-factor authentication (MFA), and identity federation
- KU8.** enterprise monitoring and alerting systems for SLA management and regulatory compliance tracking
- KU9.** key global and local regulatory compliance standards applicable to data storage, transfer, and processing
- KU10.** incident response, risk mitigation, and continuous compliance improvement strategies

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** follow SOPs, compliance checklists, and service level agreements accurately
- GS2.** work collaboratively across cross-functional IT, compliance, and business teams
- GS3.** demonstrate autonomy in managing cloud governance tasks while aligning with organizational goals
- GS4.** interpret and communicate the business impact of Cloud Engineering decisions to stakeholders
- GS5.** perform accurate reviews of technical documents, policies, and audit logs to ensure quality and completeness
- GS6.** exercise sound judgment in identifying potential compliance gaps and prioritizing corrective actions
- GS7.** provide structured feedback on proposed policies, architecture models, and compliance issues
- GS8.** organize and prioritize workload to meet critical regulatory deadlines and IT project milestones
- GS9.** use problem-solving and root-cause analysis tools to resolve security, compliance, and performance issues

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. analyze organizational business strategies, goals, and IT infrastructure requirements to determine cloud adoption needs	3	5	-	2
PC2. identify and classify all cloud-deployed and on-premise assets based on sensitivity, criticality, and compliance impact	3	4	-	1
PC3. assess integration and interoperability of hybrid/multi-cloud and on-premise environments with a focus on data flow, latency, and security	3	5	-	2
PC4. define programming, infrastructure, and design standards for developing, deploying, and maintaining scalable and secure cloud solutions	3	5	-	2
PC5. establish enterprise-wide standards for provisioning, scaling, and decommissioning of cloud resources	3	5	-	2
PC6. develop role-based access control (RBAC) policies and identity management frameworks for cloud environments	3	4	-	2
PC7. design performance monitoring standards, incident management workflows, and SLAs for cloud infrastructure	3	5	-	2
PC8. create guidelines for contributing to and managing open-source software, as relevant to the enterprise's cloud operations	3	4	-	2
PC9. identify applicable legal, regulatory, and industry-specific standards (e.g., PCI DSS, HIPAA, ISO 27001, GDPR, SOC 2)	2	4	-	2
PC10. implement governance frameworks and internal controls to ensure compliance with defined policies and regulatory standards	2	5	-	2
PC11. monitor, audit, and continuously improve compliance adherence using automated tools and manual checkpoints	2	4	-	1

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8305
NOS Name	Define enterprise cloud guidelines and implement regulatory compliances
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Cloud Computing
NSQF Level	6.5
Credits	2
Version	3.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

SSC/N8304: Assess and screen cloud vendors and third-party service providers

Description

This unit is about identifying leading cloud vendors and third-party cloud service providers and evaluating their technologies to select the most appropriate technology provider.

Scope

The scope covers the following :

- Assess and screen cloud vendors and third-party service providers

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** develop criteria and methods for shortlisting potential cloud vendors based on business and technical requirements
- PC2.** gather, analyze, and compare service offerings, including pricing, performance benchmarks, SLAs, and support models
- PC3.** evaluate the performance history, SLA compliance, market reputation, and customer feedback of cloud vendors
- PC4.** design and apply a structured framework to assess and periodically review existing cloud vendors and service providers
- PC5.** assess vendor lock-in risks and long-term dependencies to ensure operational flexibility and business continuity
- PC6.** evaluate security risks, data privacy implications, and compliance posture of vendors against required standards (e.g., GDPR, ISO/IEC 27001, HIPAA)
- PC7.** identify financial, legal, and operational risks associated with third-party cloud partnerships
- PC8.** evaluate and align vendor business continuity and disaster recovery plans with organizational needs
- PC9.** develop contingency strategies including backup vendors, exit plans, and data migration procedures in case of vendor failure or non-compliance
- PC10.** define and formalize partnership models, SLAs, operational procedures, and legal contracts for vendor collaboration
- PC11.** create a multi-vendor cloud ecosystem that ensures resilience, interoperability, and risk mitigation

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** policies and protocols for sourcing, onboarding, and managing third-party service providers
- KU2.** guidelines related to vendor risk management, data sharing, and compliance documentation

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- KU3.** stakeholders and functions to be involved during vendor evaluation and selection
- KU4.** industry-standard tools, templates, and evaluation frameworks used for cloud vendor assessment
- KU5.** key technical and business features of cloud services, including architecture requirements, pricing models, scalability, and support
- KU6.** emerging trends and innovations in cloud computing (e.g., serverless, AI integration, green cloud practices)
- KU7.** methods to assess vendor dependency and lock-in scenarios
- KU8.** national and international regulatory standards and their relevance to cloud vendors
- KU9.** types of vendor contracts, including licensing, subscription, and managed service agreements
- KU10.** approaches to defining effective partnership models and governance structures
- KU11.** how to develop and maintain contingency and business continuity plans for critical cloud services

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** follow organizational procedures, service level agreements, and compliance frameworks
- GS2.** communicate effectively with technical and business stakeholders
- GS3.** apply structured decision-making and prioritization techniques
- GS4.** plan, schedule, and organize tasks to meet evaluation timelines
- GS5.** review and validate the completeness and accuracy of assessments
- GS6.** seek expert input when dealing with complex or unclear vendor situations
- GS7.** interpret vendor documentation and convert findings into actionable insights
- GS8.** provide clear, evidence-based feedback and recommendations
- GS9.** work both independently and collaboratively in vendor evaluation teams
- GS10.** contribute to the continuous improvement of the vendor selection and management process

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. develop criteria and methods for shortlisting potential cloud vendors based on business and technical requirements	3	5	-	2
PC2. gather, analyze, and compare service offerings, including pricing, performance benchmarks, SLAs, and support models	3	4	-	1
PC3. evaluate the performance history, SLA compliance, market reputation, and customer feedback of cloud vendors	3	5	-	2
PC4. design and apply a structured framework to assess and periodically review existing cloud vendors and service providers	3	5	-	2
PC5. assess vendor lock-in risks and long-term dependencies to ensure operational flexibility and business continuity	3	5	-	2
PC6. evaluate security risks, data privacy implications, and compliance posture of vendors against required standards (e.g., GDPR, ISO/IEC 27001, HIPAA)	3	4	-	2
PC7. identify financial, legal, and operational risks associated with third-party cloud partnerships	3	5	-	2
PC8. evaluate and align vendor business continuity and disaster recovery plans with organizational needs	3	4	-	2
PC9. develop contingency strategies including backup vendors, exit plans, and data migration procedures in case of vendor failure or non-compliance	2	4	-	2
PC10. define and formalize partnership models, SLAs, operational procedures, and legal contracts for vendor collaboration	2	5	-	2

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. create a multi-vendor cloud ecosystem that ensures resilience, interoperability, and risk mitigation	2	4	-	1
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8304
NOS Name	Assess and screen cloud vendors and third-party service providers
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Cloud Computing
NSQF Level	6.5
Credits	2
Version	3.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

Qualification Pack

PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

Qualification Pack

- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

The assessment will consist of a blend of hands-on practical evaluations, viva-voce, and online proctored scenario-based multiple-choice questions ensuring a thorough evaluation of the individual's proficiency in learning outcomes, practical understanding, and real-world application of concepts.

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

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National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SSC/N8312.Assist in creating proposals and cost models for implementing cloud solutions	30	50	0	20	100	10
SSC/N8311.Validate project plans, costs, and resources required for implementing cloud solutions	30	50	0	20	100	10
SSC/N8310.Define parameters and tools to monitor and manage cloud services	30	50	0	20	100	10
SSC/N8309.Ensure cloud platforms and services are scalable, resilient, and fault-tolerant	30	50	0	20	100	10
SSC/N8308.Design and maintain strong security architecture	30	50	0	20	100	10
SSC/N8307.Analyze, design, and maintain Cloud Engineerure that meets business requirements	30	50	0	20	100	10
SSC/N8306.Translate business-level use cases into technical requirements and specifications	30	50	0	20	100	10
SSC/N8305.Define enterprise cloud guidelines and implement regulatory compliances	30	50	0	20	100	10
SSC/N8304.Assess and screen cloud vendors and third-party service providers	30	50	0	20	100	10
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	10
Total	290	480	-	180	950	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

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Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.